



西南交通大学
Southwest Jiaotong University

Matrix Analysis

Avik Ranjan Adhikary

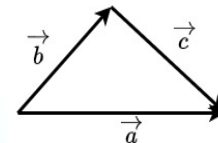
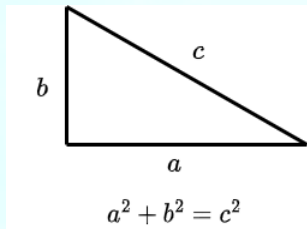
Dept. of Mathematics, SWJTU, China.

Sept 27, 2026



Inner Product & Orthogonality

Pythagoras' Theorem



The length of a vector is called by **norm**, denoted by $\|\vec{a}\|$

The length of $\|\vec{a}\| = \sqrt{a_1^2 + a_2^2}$

$$\|\vec{b}\| = \sqrt{b_1^2 + b_2^2}$$

$$\vec{c} = \vec{a} - \vec{b},$$

$$\|\vec{c}\| = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2}$$

If the vectors $\vec{a}$, and $\vec{b}$ are to be orthogonal, then

$$\|\vec{a}\|^2 + \|\vec{b}\|^2 - \|\vec{c}\|^2 = 0.$$

We have,

$$\|\vec{a}\|^2 + \|\vec{b}\|^2 - \|\vec{c}\|^2 = 2(a_1b_1 + a_2b_2).$$

Hence $\vec{a}$ and $\vec{b}$ are ORTHOGONAL to each other or not, depends on $a_1b_1 + a_2b_2 = 0$ or not.

Let us consider this as a function $f(\vec{a}, \vec{b}) = a_1b_1 + a_2b_2$.

- ▶ For $\mathbb{R}^3$, $f(\vec{a}, \vec{b}) = a_1b_1 + a_2b_2 + a_3b_3$.
- ▶ For $\mathbb{R}^n$, $f(\vec{a}, \vec{b}) = a_1b_1 + a_2b_2 + \cdots + a_nb_n$.

In $\mathbb{R}^3$, $f(\vec{a}, \vec{b}) = a_1b_1 + a_2b_2 + a_3b_3$

- ▶ $f(\vec{a}, \vec{b}) = f(\vec{b}, \vec{a})$.
- ▶ $f(\alpha \vec{a}, \vec{b}) = \alpha f(\vec{a}, \vec{b})$
- ▶ If we replace $\vec{a}$ by $\vec{c} + \vec{d}$, then $f(\vec{c} + \vec{d}, \vec{b}) = f(\vec{c}, \vec{b}) + f(\vec{d}, \vec{b})$, since $f(\vec{a}, \vec{b}) = f(\vec{b}, \vec{a})$, $f(\vec{a}, \vec{b} + \vec{c}) = f(\vec{a}, \vec{b}) + f(\vec{a}, \vec{c})$.

Now, let us observe how these properties help.

Let $\{(1, 2, 3), (2, 3, -15), (2, 5, 7)\}$ be a basis of $\mathbb{R}^3$. Express $(3, 4, 5)$ as a linear combination of these three vectors.

$$(3, 4, 5) = a(1, 2, 3) + b(2, 3, -15) + c(2, 5, 7)$$

Then we need to solve the following system of equations

$$a + 2b + 2c = 3$$

$$2a + 3b + 5c = 4$$

$$3a - 15b + 7c = 5$$

After solving, we get $a = \frac{33}{5}$, $b = \frac{1}{10}$, $c = -\frac{19}{10}$

Example

Let us take an orthogonal basis $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ of $\mathbb{R}^3$, where $\vec{v}_1 = (1, 2, 3)$, $\vec{v}_2 = (2, -1, 0)$, and $\vec{v}_3 = (3, 6, -5)$.

$f(\vec{v}_1, \vec{v}_2) = 1 \times 2 + 2 \times (-1) + 3 \times 0 = 0$, we can check $f(\vec{v}_1, \vec{v}_3)$, $f(\vec{v}_2, \vec{v}_3)$ also. We can see that $\vec{v}_1, \vec{v}_2, \vec{v}_3$ are orthogonal.

Now let us write $\vec{v} = (3, 4, 5)$ as a linear combinations of $\vec{v}_1, \vec{v}_2, \vec{v}_3$.

$$\vec{v} = a\vec{v}_1 + b\vec{v}_2 + c\vec{v}_3$$

Example

$$\begin{aligned} f(\vec{v}_1, \vec{v}) &= f(\vec{v}_1, a\vec{v}_1 + b\vec{v}_2 + c\vec{v}_3) \\ &= f(\vec{v}_1, a\vec{v}_1) + f(\vec{v}_1, b\vec{v}_2) + f(\vec{v}_1, c\vec{v}_3) \\ &= af(\vec{v}_1, \vec{v}_1) + bf(\vec{v}_1, \vec{v}_2) + cf(\vec{v}_1, \vec{v}_3) \\ &= af(\vec{v}_1, \vec{v}_1) \end{aligned}$$

Hence, $a = \frac{f(\vec{v}_1, \vec{v})}{f(\vec{v}_1, \vec{v}_1)}$.

$f(\vec{v}_1, \vec{v}_1)$ is the square of the length of $\vec{v}_1$. $f(\vec{v}_1, \vec{v}_1) \geq 0$. It is zero only when $\vec{v}_1 = 0$. However, since $\vec{v}_1$ is a basis here, so $\vec{v}_1 \neq 0$.

Similarly, $b = \frac{f(\vec{v}_2, \vec{v})}{f(\vec{v}_2, \vec{v}_2)}$, $c = \frac{f(\vec{v}_3, \vec{v})}{f(\vec{v}_3, \vec{v}_3)}$.

Another Advantage

Earlier we solved a problem to express $(3, 4, 5)$ as a linear combination of $(1, 2, 3)$, $(2, 3, -15)$, $(2, 5, 7)$. We got

$$a = \frac{33}{5}, \quad b = \frac{1}{10}, \quad c = -\frac{19}{10}$$

If we change the previous basis, just by changing the last vector of the basis, i.e., let the new basis be $(1, 2, 3)$, $(2, 3, -15)$, $(2, 5, 17)$, now if we want to express $(3, 4, 5)$ as the linear combination of these vectors, then

$$a = \frac{71}{5}, \quad b = -\frac{9}{5}, \quad c = -\frac{19}{5}$$

If we just change one vector in the basis, the entire combination of a , b , c changes.

Another Advantage

Now consider that we work with an orthogonal basis $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$. Let us write $\vec{v}$ as a linear combination of $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$.

$$\vec{v} = \alpha_1 \vec{v}_1 + \alpha_2 \vec{v}_2 + \dots + \alpha_n \vec{v}_n$$

As we have seen earlier, we can derive $\alpha_i = \frac{f(\vec{v}_i, \vec{v})}{f(\vec{v}_i, \vec{v}_i)}$, it solely depends on $\vec{v}_i$ and $\vec{v}$.

So, if we replace, let say v_n with another orthogonal vector $\vec{u}$, only α_n will change, the other coefficients will remain same.

The function f is called **inner product**, if it has the following properties

1. $\forall \vec{x}_1, \vec{x}_2, \vec{y} \in V, \forall \alpha, \beta \in \mathbb{R}, f(\alpha \vec{x}_1 + \beta \vec{x}_2, \vec{y}) = \alpha f(\vec{x}_1, \vec{y}) + \beta f(\vec{x}_2, \vec{y})$.
2. $\forall \vec{x} \in V, f(\vec{x}, \vec{x}) \geq 0$, and $f(\vec{x}, \vec{x}) = 0 \implies \vec{x} = 0$.
3. $\forall \vec{x}, \vec{y} \in V, f(\vec{x}, \vec{y}) = f(\vec{y}, \vec{x})$

If V is a real vector space, then we denote $f(\vec{u}, \vec{v})$ as $\langle \vec{u}, \vec{v} \rangle$.

If V is a complex vector space $\mathbb{C}$, then $f(\vec{u}, \vec{v}) = \overline{f(\vec{v}, \vec{u})}$, since in $\mathbb{R}$, $|x| = \sqrt{x^2}$, however in $\mathbb{C}$, $|z| = \sqrt{z\bar{z}}$.

Example

In $\mathbb{R}^3$, let $\vec{u} = (u_1, u_2, u_3)$ and $\vec{v} = (v_1, v_2, v_3)$, and $f(\vec{u}, \vec{v}) = |u_1 v_1 + u_2 v_2 + u_3 v_3|$. Does f define a real inner product?

Let us take a counterexample to disprove it. Let $\vec{u} = \vec{v} = (1, 1, 1)$. Then show $f(-2\vec{u}, \vec{v}) \neq -2f(\vec{u}, \vec{v})$.

Real inner product space is Euclidean Space.

Orthogonal & Orthonormal Set



Orthogonal Set

A subset A of an inner product space V is called orthogonal if

$$\forall \vec{u} \neq \vec{v} \in A, \langle \vec{u}, \vec{v} \rangle = 0.$$

Orthonormal Set

A subset A of an inner product space V is called orthonormal if

$$\forall \vec{v} \in A, \|\vec{v}\| = 1,$$

$$\forall \vec{u} \neq \vec{v} \in A, \langle \vec{u}, \vec{v} \rangle = 0.$$

In an inner product space, if the vectors in the basis are mutually orthogonal, then we call it orthogonal basis. Similarly, in the basis if the vectors are mutually orthonormal, we call it orthonormal basis.

Orthogonal Vectors are Linearly Independent

Remember: **Must be non-null vectors**. Because if it is null vector, then $\langle \vec{v}, \vec{0} \rangle = 0$, for all $\vec{v}$, trivial.

Consider n orthogonal vectors, $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$, need to show if

$$\sum_{i=1}^n a_i \vec{v}_i = \vec{0},$$

then $\forall i \in \{1, 2, \dots, n\}$, $a_i = 0$.

Take any $k \in \{1, 2, \dots, n\}$,

$$\left\langle \sum_{i=1}^n a_i \vec{v}_i, \vec{v}_k \right\rangle = \langle \vec{0}, \vec{v}_k \rangle = 0.$$

Expanding, we get

$$a_1 \langle \vec{v}_1, \vec{v}_k \rangle + a_2 \langle \vec{v}_2, \vec{v}_k \rangle + \cdots + a_k \langle \vec{v}_k, \vec{v}_k \rangle + \cdots + a_n \langle \vec{v}_n, \vec{v}_k \rangle = 0$$

since $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ are orthogonal, hence $\langle \vec{v}_i, \vec{v}_k \rangle = 0$ for $i \neq k$. Now, we have

$$a_k \langle \vec{v}_k, \vec{v}_k \rangle = 0$$

$\vec{v}_k \neq 0$, therefore $\langle \vec{v}_k, \vec{v}_k \rangle \neq 0$.

Therefore $a_k = 0$.

Since $k \in \{1, 2, 3, \dots, n\}$ was arbitrary, we have $a_1 = a_2 = \cdots = a_n = 0$.

Some important relations in REAL inner product

1. $\|\vec{v}\|^2 = \langle \vec{v}, \vec{v} \rangle$.
2. $\|c\vec{v}\| = |c|\|\vec{v}\|$.

$$\begin{aligned}\|c\vec{v}\| &= \sqrt{\langle c\vec{v}, c\vec{v} \rangle} \\ &= \sqrt{c\langle \vec{v}, c\vec{v} \rangle} \\ &= \sqrt{c^2\langle \vec{v}, \vec{v} \rangle} \\ &= |c|\sqrt{\langle \vec{v}, \vec{v} \rangle} \\ &= |c|\|\vec{v}\|\end{aligned}$$

Some important relations in REAL inner product space



For any two vectors $\vec{u}$, $\vec{v}$ in a real inner product space, $\langle \vec{u}, \vec{v} \rangle = 0$, if and only if

$$\|\vec{u} + \vec{v}\| = \|\vec{u} - \vec{v}\|$$

First, let us assume, $\langle \vec{u}, \vec{v} \rangle = 0$, then

$$\begin{aligned}\|\vec{u} + \vec{v}\| &= \sqrt{\langle \vec{u} + \vec{v}, \vec{u} + \vec{v} \rangle} \\ &= \sqrt{\langle \vec{u}, \vec{u} + \vec{v} \rangle + \langle \vec{v}, \vec{u} + \vec{v} \rangle} \\ &= \sqrt{\langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle} \\ &= \sqrt{\langle \vec{u}, \vec{u} \rangle + 2\langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{v} \rangle} \\ &= \sqrt{\langle \vec{u}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle}\end{aligned}$$

Some important relations in REAL inner product

Similarly, we can show

$$\begin{aligned}\|\vec{u} - \vec{v}\| &= \sqrt{\langle \vec{u}, \vec{u} \rangle - 2\langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{v} \rangle} \\ &= \sqrt{\langle \vec{u}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle}\end{aligned}$$

Therefore

$$\|\vec{u} + \vec{v}\| = \|\vec{u} - \vec{v}\|$$

Now the other way, given $\|\vec{u} + \vec{v}\| = \|\vec{u} - \vec{v}\|$, need to show $\langle \vec{u}, \vec{v} \rangle = 0$

$$\begin{aligned}\|\vec{u} + \vec{v}\| &= \|\vec{u} - \vec{v}\| \\ (\|\vec{u} + \vec{v}\|)^2 &= (\|\vec{u} - \vec{v}\|)^2 \\ \langle \vec{u}, \vec{v} \rangle &= -\langle \vec{u}, \vec{v} \rangle \implies \langle \vec{u}, \vec{v} \rangle = 0.\end{aligned}$$

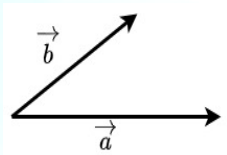
Orthogonal Matrix

A real matrix A of order n , is said to be orthogonal if

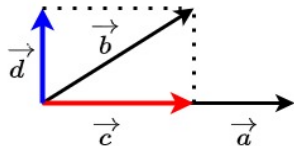
- ▶ $AA^T = A^T A = I_n$.
- ▶ The rows of the matrix will form an orthonormal basis.
- ▶ The columns of the matrix will form an orthonormal basis.

Component of a vector along another vector

Let us consider two vectors $\vec{a}$, $\vec{b}$ as follows:



can we write $\vec{b}$ as sum of two vectors $\vec{c}$ and $\vec{d}$, such that $\vec{c}$ is along the vector $\vec{a}$ and $\vec{d}$ is perpendicular to $\vec{a}$. $\vec{c}$ is called component of $\vec{b}$ along $\vec{a}$.



Component of a vector along another vector



So we have $\vec{b} = \vec{c} + \vec{d}$, where we want $\vec{c}$ along $\vec{a}$ and $\vec{d}$ perpendicular to $\vec{a}$.

$$\begin{aligned}\langle \vec{a}, \vec{b} \rangle &= \langle \vec{a}, \vec{c} + \vec{d} \rangle \\ &= \langle \vec{a}, \vec{c} \rangle + \langle \vec{a}, \vec{d} \rangle\end{aligned}$$

since, $\vec{a}$, and $\vec{d}$ are perpendicular $\langle \vec{a}, \vec{d} \rangle = 0$. Since $\vec{c}$ is along $\vec{a}$, therefore $\vec{c} = \lambda \vec{a}$. Hence,

$$\begin{aligned}\langle \vec{a}, \vec{b} \rangle &= \langle \vec{a}, \lambda \vec{a} \rangle \\ &= \lambda \langle \vec{a}, \vec{a} \rangle\end{aligned}$$

Hence $\lambda = \frac{\langle \vec{a}, \vec{b} \rangle}{\langle \vec{a}, \vec{a} \rangle}$, since $\vec{a} \neq \vec{0}$. Therefore $\vec{c} = \lambda \vec{a} = \frac{\langle \vec{a}, \vec{b} \rangle}{\langle \vec{a}, \vec{a} \rangle} \vec{a}$ is the component of $\vec{b}$ along $\vec{a}$.

Gram-Schmidt Process

Note that if we remove $\vec{c}$, the component of $\vec{b}$ along $\vec{a}$, from $\vec{b}$, then we get $\vec{d}$, and $\vec{d}$ is orthogonal to $\vec{a}$.

Let us start with two vectors $\{\vec{a}, \vec{b}\}$, which may not be orthogonal to each other. Compute

$$\vec{d} = \vec{b} - \frac{\langle \vec{a}, \vec{b} \rangle}{\langle \vec{a}, \vec{a} \rangle} \vec{a}$$

Then, $\vec{d}$ and $\vec{a}$ are orthogonal to each other.

Gram Schmidt Process

Let $\{\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n\}$ be a linearly independent set of vectors. Let, for $k = 1, 2, \dots, n$,

$$\vec{v}_k = \vec{u}_k - \sum_{i=1}^{k-1} \frac{\langle \vec{u}_k, \vec{v}_i \rangle}{\langle \vec{v}_i, \vec{v}_i \rangle} \vec{v}_i$$

and

$$\vec{w}_k = \frac{\vec{v}_k}{\|\vec{v}_k\|}$$

Then $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ is an orthogonal basis, and $\{\vec{w}_1, \vec{w}_2, \dots, \vec{w}_n\}$ is an orthonormal basis.

Example

Apply Gram Schmidt Process to the vectors $(1, 0, 1)$, $(1, 0, -1)$ and $(1, 3, 4)$, to obtain an orthonormal basis for $\mathbb{R}^3$, with the standard inner product.

Let

$$\vec{u}_1 = (1, 0, 1)$$

$$\vec{u}_2 = (1, 0, -1)$$

$$\vec{u}_3 = (1, 3, 4)$$

Now first take $\vec{v}_1 = \vec{u}_1$.

Next, $\vec{v}_2 = \vec{u}_2 - \frac{\langle \vec{u}_2, \vec{v}_1 \rangle}{\langle \vec{v}_1, \vec{v}_1 \rangle} \vec{v}_1$, $\vec{v}_3 = \vec{u}_3 - \frac{\langle \vec{u}_3, \vec{v}_1 \rangle}{\langle \vec{v}_1, \vec{v}_1 \rangle} \vec{v}_1 - \frac{\langle \vec{u}_3, \vec{v}_2 \rangle}{\langle \vec{v}_2, \vec{v}_2 \rangle} \vec{v}_2$

Gram Schmidt Process

$$\langle \vec{u}_2, \vec{v}_1 \rangle = 0$$

Therefore, $\vec{v}_2 = (1, 0, -1)$.

Now, $\langle \vec{u}_3, \vec{v}_1 \rangle = 5$, $\langle \vec{u}_3, \vec{v}_2 \rangle = -3$, $\langle \vec{v}_1, \vec{v}_1 \rangle = 2$, $\langle \vec{v}_2, \vec{v}_2 \rangle = 2$

Therefore, $\vec{v}_3 = (1, 3, 4) - \frac{5}{2}(1, 0, 1) + \frac{3}{2}(1, 0, -1) = (0, 3, 0)$

Hence the orthogonal basis is

$$\{(1, 0, 1), (1, 0, -1), (0, 3, 0)\}$$

Corresponding orthonormal vectors are

$$\vec{w}_1 = \frac{\vec{v}_1}{\|\vec{v}_1\|} = \frac{1}{\sqrt{2}}(1, 0, 1)$$

$$\vec{w}_2 = \frac{\vec{v}_2}{\|\vec{v}_2\|} = \frac{1}{\sqrt{2}}(1, 0, -1)$$

$$\vec{w}_3 = \frac{\vec{v}_3}{\|\vec{v}_3\|} = \frac{1}{3}(0, 3, 0) = (0, 1, 0)$$

QR Factorization

A square matrix A can be decomposed as $A=QR$.

Q: Orthogonal Matrix

R: Upper triangular Matrix

Working Procedure:

- ▶ Consider the columns of A as the vectors, and then perform Gram Schmidt Process, to generate the orthonormal vectors.
- ▶ The orthonormal vectors corresponding to the columns of A , constructs Q .
- ▶ In R we record the steps of the Gram-Schmidt process.

QR Factorization

In the earlier example, consider the vectors as the columns of A , i.e.,

$$A = [\vec{u}_1 | \vec{u}_2 | \vec{u}_3] = \underbrace{\left[\frac{\vec{v}_1}{\|\vec{v}_1\|} \mid \frac{\vec{v}_2}{\|\vec{v}_2\|} \mid \frac{\vec{v}_3}{\|\vec{v}_3\|} \right]}_Q \underbrace{\begin{bmatrix} \|\vec{v}_1\| & \frac{\langle \vec{u}_2, \vec{v}_1 \rangle}{\langle \vec{v}_1, \vec{v}_1 \rangle} \|\vec{v}_1\| & \frac{\langle \vec{u}_3, \vec{v}_1 \rangle}{\langle \vec{v}_1, \vec{v}_1 \rangle} \|\vec{v}_1\| \\ 0 & \|\vec{v}_2\| & \frac{\langle \vec{u}_3, \vec{v}_2 \rangle}{\langle \vec{v}_2, \vec{v}_2 \rangle} \|\vec{v}_2\| \\ 0 & 0 & \|\vec{v}_3\| \end{bmatrix}}_R$$

$$\vec{u}_1 = \vec{v}_1$$

$$\vec{u}_2 = \vec{v}_2 + \frac{\langle \vec{u}_2, \vec{v}_1 \rangle}{\langle \vec{v}_1, \vec{v}_1 \rangle} \vec{v}_1$$

$$\vec{u}_3 = \vec{v}_3 + \frac{\langle \vec{u}_3, \vec{v}_2 \rangle}{\langle \vec{v}_2, \vec{v}_2 \rangle} \vec{v}_2 + \frac{\langle \vec{u}_3, \vec{v}_1 \rangle}{\langle \vec{v}_1, \vec{v}_1 \rangle} \vec{v}_1$$

QR Factorization

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 3 \\ 1 & -1 & 4 \end{bmatrix} = \underbrace{\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \end{bmatrix}}_Q \underbrace{\begin{bmatrix} \sqrt{2} & 0 & \frac{5\sqrt{2}}{2} \\ 0 & \sqrt{2} & -\frac{3\sqrt{2}}{2} \\ 0 & 0 & 3 \end{bmatrix}}_R$$

In general

$$Q = \left[\frac{\vec{v}_1}{\|\vec{v}_1\|} \quad \frac{\vec{v}_2}{\|\vec{v}_2\|} \quad \cdots \quad \frac{\vec{v}_n}{\|\vec{v}_n\|} \right]$$

$$R = [r_{ij}]_{i,j=1}^n, \text{ where } r_{ij} = \begin{cases} \|\vec{v}_i\| & \text{when } i = j \\ \frac{\langle \vec{v}_j, \vec{v}_i \rangle}{\langle \vec{v}_i, \vec{v}_i \rangle} \|\vec{v}_i\| & \text{when } j > i \\ 0 & \text{when } i < j \end{cases}$$

Student ID: 2026220088, 2026220104, 2026220117